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Palaeomagnetism of Australian Hematite Ore Bodies—I The Middleback Ranges of South Australia

F. H. Chamalaun and H. Porath

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Summary

The results of a palaeomagnetic study on high-grade hematite ore bodies of South Australia are reported. The two major ore bodies, Iron Monarch and Iron Prince, gave consistent directions of magnetization with either steep to moderate negative and positive inclinations. The positive and negative groups of Iron Monarch are not antiparallel. Because of the presumably long time interval represented by the sampling, this is interpreted as being due to polar wandering with a superimposed reversal of the Earth's field. The remanent magnetization is carried by hematite and is extremely stable against alternating field and thermal demagnetization. The intensity of the NRM is very weak compared to the TRM, which suggests it to be a CRM in origin. Lack of a detailed polar wandering curve for the Pre-Cambrian of Australia prevents precise palaeomagnetic dating of the ores, but the results clearly suggest a Pre-Cambrian age. The magnetic anisotropy of the ores is low and the distribution of the principal susceptibility axes reflects in part the structure of the host rock. The Iron Duke orebody did not yield a reliable palaeomagnetic result.

1. Introduction

High-grade hematite ore bodies associated with banded iron formations occur in the Pre-Cambrian shield areas of all continents and represent the world's most important source of iron ore. Since they have many features in common, they are referred to as Lake Superior type ore bodies. Numerous hypotheses have been advanced to account for their origin, but, as Park (1959) has pointed out, great confusion exists in the interpretation of the field evidence and has impeded the emergence of a really satisfactory hypothesis.

Because of the obvious genetic relationship between ores and banded iron formation, a replacement origin is now generally accepted as being responsible for the ore formation. The absence of slumping indicates that the ores did not form exclusively from leaching of silica from the iron formation and compaction of the remaining iron, but that in addition some enrichment of iron must have taken place. Little is known in detail about the physical chemistry of this process. The main problem is the nature and source of the fluids that transported the enormous volumes of silica and iron.

Miles (1954) and Dorr (1965) proposed that hot fluids leached the silica from the iron formations and introduced iron leached from adjacent parts of the iron formation. Dorr advocates water at or near supercritical temperatures as the main

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transporting agent. The structure of the iron formations is thought to have controlled the movement of the fluids, either by acting as channels (Dorr 1965) or by forming structural traps (Miles 1954) for the ore forming solutions. In the case of the Middleback Ranges, Miles suggests the fluids to be associated with granitic intrusions.

A further problem is the determination of the age of the ores. As the ore bodies usually occur in isolated outcrops, they cannot be placed in a definite stratigraphical sequence. However, in most cases a Pre-Cambrian age is assumed. Since hematite is magnetically extremely stable, one may expect the ores to yield palaeomagnetic pole positions, which, when compared with the polar wandering curve of the continent, should enable one to determine the age of the ore palaeomagnetically.

The potential usefulness of magnetic studies of Lake Superior type ore bodies was demonstrated by Gross & Strangway (1961). They investigated the remanent magnetization of two ore bodies in Mauritania, North Africa and showed that one ore body was magnetized in the Pre-Cambrian while the formation was still flat lying, whereas the second one was found to be formed by surface leaching in recent times. Similar studies were recently carried out by Symons (1966, 1967) on the Lake Superior ores of North America.

In Australia a great number of Lake Superior type ore bodies have been discovered, particularly in South Australia and Western Australia. A systematic study of the major and most accessible ore bodies has now been carried out (Porath 1967). This paper will deal with the results obtained for the ore deposits of the Middleback Ranges, South Australia, whereas the Western Australian results will be described in a subsequent paper.

2. Geological setting

The geology of the Middleback Ranges has been described by Miles (1954). They form a set of subparallel but discontinuous north–south trending ridges some 20 miles west of Whyalla on Eyre Peninsula (Fig. 1). The ore bodies lie generally in pitching synclines and anticlines within the Middleback Group, which consists of banded iron formation, or jaspilites, schists, dolomite and sericitic quartzites. The Middleback Group overlies a gneissic and granitic basement complex and is in turn overlain by the Moonabie Group of conglomerates and breccias. These are followed by the Corunna Group and Tertiary deposits.

Compston *et al.* (1966) have dated the metamorphism of the basement gneiss complex at 1800 m.y. and an illite member of the Corunna Group at 1535 m.y. $\pm$ 20 m.y. Pebbles of jaspilites have been found in the Corunna Conglomerates (Miles 1954), which is intruded by the Gawler Range Porphyry. The latter has been dated by Compston *et al.* (1966) at 1535 m.y. $\pm$ 25 m.y. and this, therefore, sets the minimum age of the Middleback Group.

No pebbles of hematite ore have been found in the Corunna Conglomerate (Whitten, personal communication). It appears, therefore, that the ores are younger than 1535 m.y., unless they were not exposed at the time of the deposition of the conglomerate. However, the fact that the ores show no sign of metamorphism suggests them to be younger than 1800 m.y., which is the metamorphic age of the gneiss complex.

Of the numerous hematite ore bodies of the area only Iron Monarch and Iron Prince are so extensively mined that a substantial collection of samples could be obtained. The collection of the Iron Duke ore body was made from exploratory tunnels.

3. Palaeomagnetic results

(a) *Jaspilites.* The term jaspilite is used by Miles (1954) to describe the banded iron formations of the Middleback Group. They consist essentially of alternating

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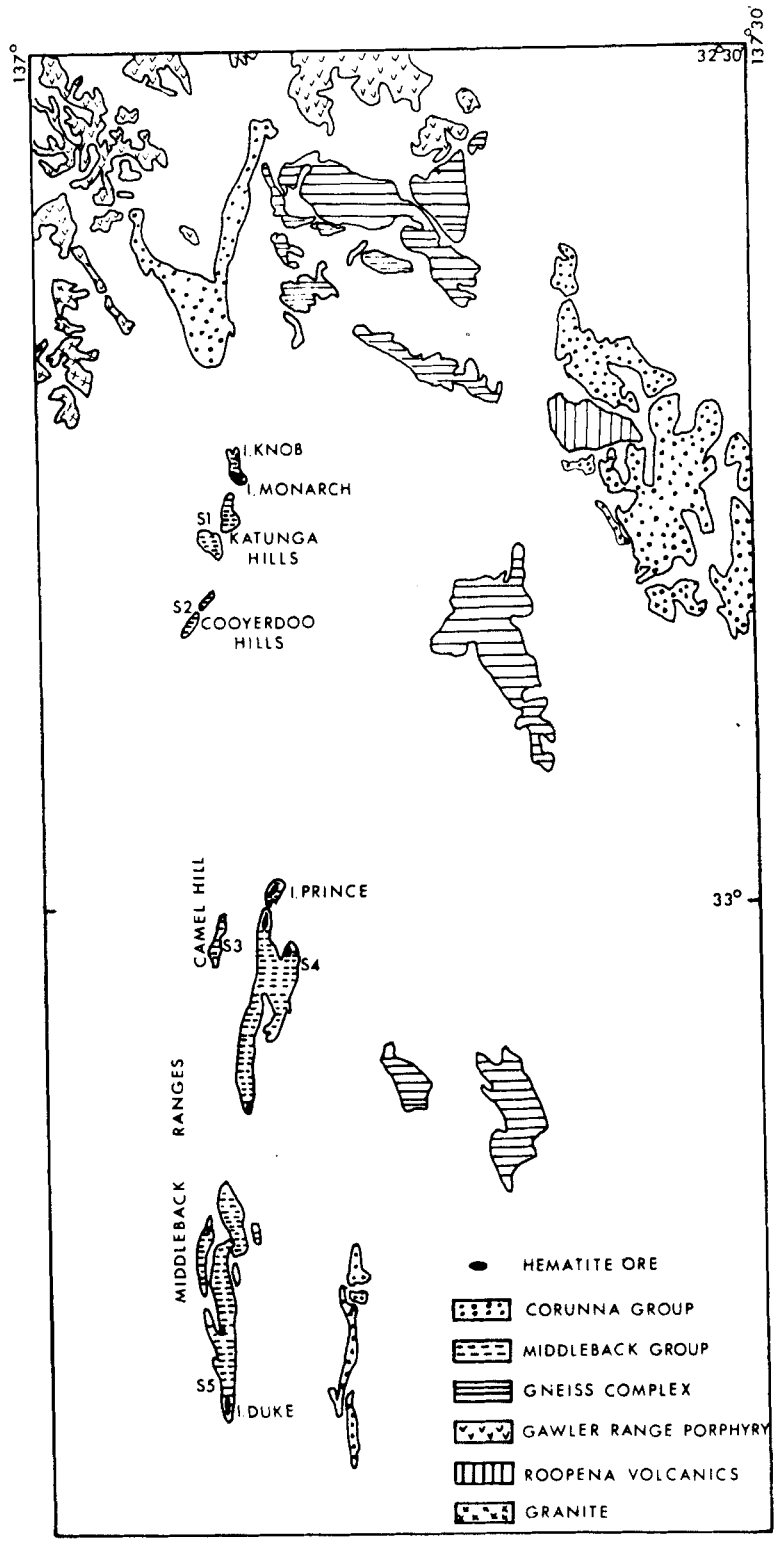


FIG. 1. Geological sketch map of the Middleback Ranges.

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layers of iron oxides and silica, varying in thickness from a few millimeters to several centimeters. The iron oxide is mainly martite in near surface outcrops, but deep drilling has shown that below the zone of oxidation the dominant mineral is magnetite. The jaspilites are thought to have been chemically precipitated in a narrow coastal basin (Miles 1954).

Forty-two samples of jaspilite were collected, mainly from surface outcrops. The directions of magnetization with respect to the present horizontal are random, as has already been shown by Mumme (1963). With respect to the banding plane the inclinations are generally low, but the declinations are still random. During alternating field and thermal demagnetization experiments changes in the directions are small and the grouping is not improved. Thermal demagnetization shows the remanence to be due to magnetite, the contribution of hematite being almost negligible. The alternating field demagnetization results indicate that the bulk of the NRM is carried by grains of low coercivities.

The anisotropy of susceptibility of the jaspilites was determined with a low field torque meter (King & Rees 1962). It is clear that the magnetic fabric will be dominated by the textural anisotropy caused by the alternation of silica and iron oxide bands. Thus the axes of minimum susceptibility form a tight group around the normal of the banding plane. Maximum susceptibility axes lie within the banding plane without any preferred orientation within this plane. Lineation and foliation vary considerably from sample to sample but their ratio is fairly constant and about 0·33.

The textural anisotropy of the jaspilites is presumably responsible for the low inclinations of the NRM directions with respect to the banding plane. The iron oxide layers can be approximated by semi-infinite magnetic sheets so that the field component perpendicular to the banding plane is reduced by $4\pi I$. Because of the high intensity of the jaspilites ($0{\cdot}1 - 1{\cdot}0 \times 10^{-2}$ e.m.u. cm^{-3}) the magnetic vector could be deflected by as much as 60 degrees.

(b) *Iron Monarch*

(i) *Ore Body.* The Iron Monarch ore body forms the largest deposit of the Middleback Ranges. It occurs in a northerly pitching syncline, terminating in the north at the North Wall amphibolite dyke, which is thought to have presented a barrier to the ore forming solutions (Miles 1954). The ore is high-grade hematite, with some admixture of manganese minerals in particular in the north-eastern parts of the body. The ore is generally hard and massive but locally grades into soft earthy hematite. Under reflected light hematite is seen as fine grained (10μ–100μ) interlocking crystals. No specular hematite is observed and magnetite is sometimes present as small needle-like inclusions in the hematite crystals.

Specimens were subjected to both thermal and magnetic cleaning techniques, in the latter case in a field of 750 oersteds peak or as in the majority of cases by heating to 600 °C and cooling in zero field. Both methods removed small components of secondary magnetization.

Under alternating field demagnetization the NRM falls by only approximately ten per cent up to 750 oersteds peak, which is not surprising since coercivities of several thousand oersteds are common in fine grained hematite (Roquet 1947, Haigh 1957). Some typical thermal demagnetization curves are shown in Fig. 2. The intensity remains virtually unchanged up to 650 °C and then drops sharply to zero. The Curie points lie between 660 °C and 680 °C. Specimens with high manganese content tend to have lower Curie points, which is similar to the observations of Haigh (1957) and Tasaki & Iida (1961), who showed that Ti, Al and Mn impurities lower the Curie point of hematite.

The directions of magnetization before and after cleaning are shown in Fig. 3. They fall into two distinct groups, one with steep positive inclinations and one with

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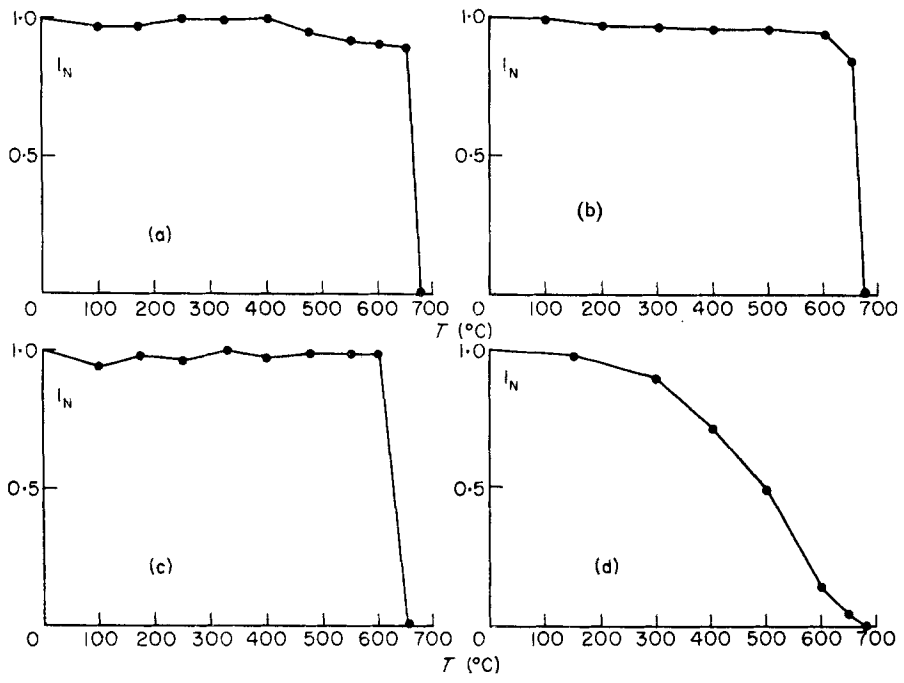


FIG. 2. Normalized thermal demagnetization curves of some typical specimens of hematite ore. (a) Iron Monarch stable hematite ore. (b) Iron Prince hematite ore. (c) Iron Monarch manganese rich ore. (d) Iron Duke magnetite rich specimens.

steep negative inclinations. The mean directions and statistical parameters are given in Table 1. Samples with unusually low intensities and intermediate directions have not been included in the computation of the mean. The declinations of the mean directions of the two groups differ by approximately 100° and we have therefore called them the positive and negative groups from Iron Monarch. Although application of the χ^2-test (Watson & Irving 1957) (Table 1) indicates that the distribution of the directions of both groups is Fisherian, the stereogram shows a somewhat elongated distribution for the positive group.

(ii) *Amphibolite dyke, intrusive dolerite dykes and contact ore.* Within the ore body several amphibolite and dolerite dykes have been found. The amphibolite dykes are extensively altered and heavily impregnated by hematite. Miles (1954)

Table 1

Mean directions of magnetization palaeomagnetic poles and statistical parameters of Iron Monarch and Iron Prince hematite ores, the significant limits in the χ^2-test are 11·07 for the distribution of declinations and 9·49 for the inclinations

		Mean direction					χ^2-test		Poles			
Ore body	*N*	*D*	*I*	*R*	*K*	α_{95}	χ^2-Dec	χ^2-Inc	Lat.	Long.	*K*	α95
I. Monarch: positive group	28	283°	+63°	25·678	12	8°	8·0	8·67	15° S	92° E	6	12°
I. Monarch: negative group	13	26°	−58	12·355	19	10°	6·07	6·85	64° N	93° W	20	9°
I. Prince	22	63°	−46	20·660	16	8°	4·18	10·9	39° N	113° W	10	9½°

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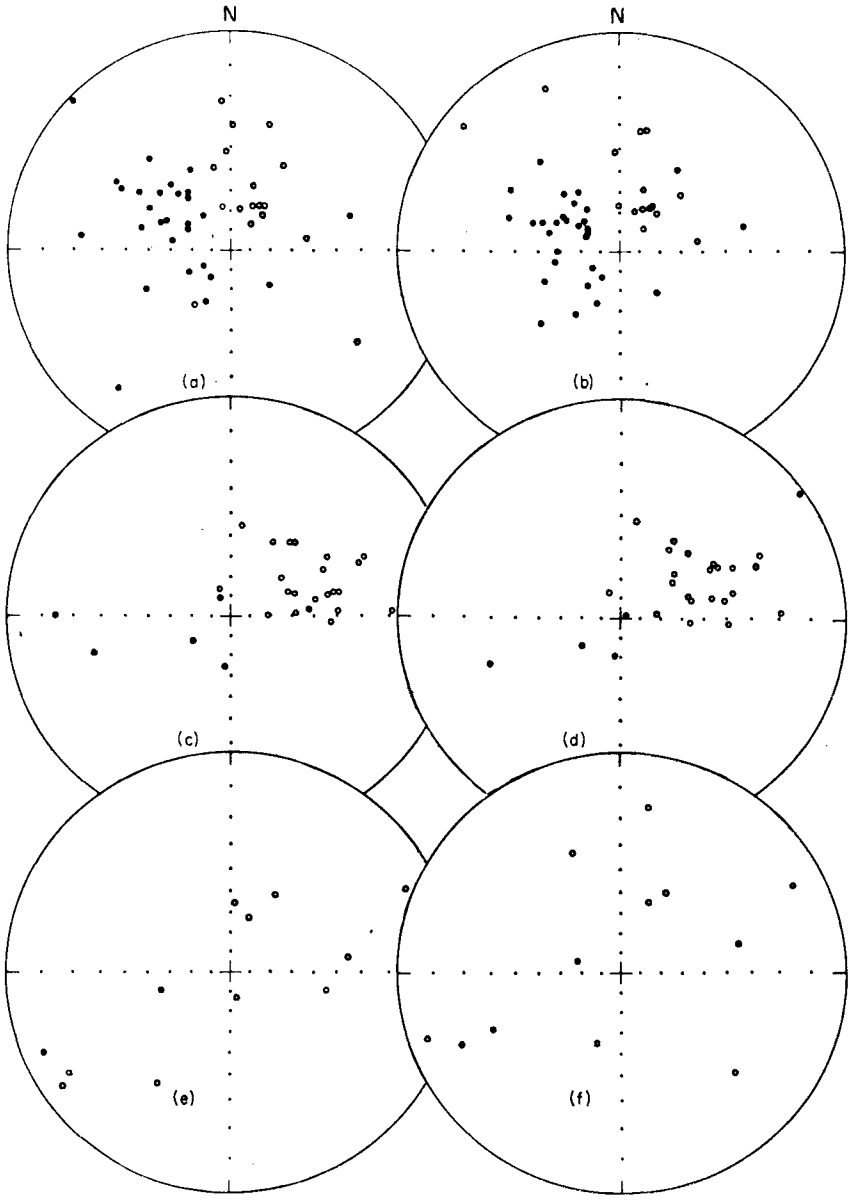


FIG. 3. Equal angle projections of the NRM directions for the Iron Monarch (a) and (b), Iron Prince (c) and (d) and Iron Duke (e) and (f) ore bodies before (left) and after cleaning (right). Closed circles indicate positive or downward directions; open circles indicate negative or upward directions.

concluded from the field relations that they were pre-ore. The age relationship between the dolerite dykes and the ore has been a matter of dispute. Jack (1922) suggested they were pre-ore, whereas Miles (1954) and Whitten (1963) maintain that they intrude the ore. We have examined polished sections of samples near the contact of the Central Dolerite Dyke and found the hematite to show lamellar twinning. Since twinning is notably absent in specimens further away from the contact, we suggest the twinning to be caused by heating of the contact ore during intrusion of the dyke and that, therefore, the dyke post-dates the ore. Thus the age

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of the dolerite dyke would determine the minimum age of the ore formation. A sample of the Central Dolerite Dyke was submitted to Dr McDougall of this department and dated at 585 m.y. by the K–Ar method. The sample was badly altered, so that argon loss had most likely occurred, and the age is therefore a minimum age.

Poor accessibility and extensive weathering made it impossible to collect more than a few samples of the Amphibolite Dyke, Central Dolerite Dyke and Contact ore and their NRM directions are shown in Fig. 4. The NRM of the North Wall Amphibolite Dyke agrees with the positive group of the ore and has similar thermal demagnetization curves. This agrees with the suggestion that the Amphibolite Dyke is pre-ore and has been impregnated by the ore forming solutions. No internally consistent directions are obtained from the samples of the Central Dolerite Dyke and the contact ore. In the case of the contact ore it is possible that the results reflect the superposition of two directions, i.e. the original ore direction and the dyke direction.

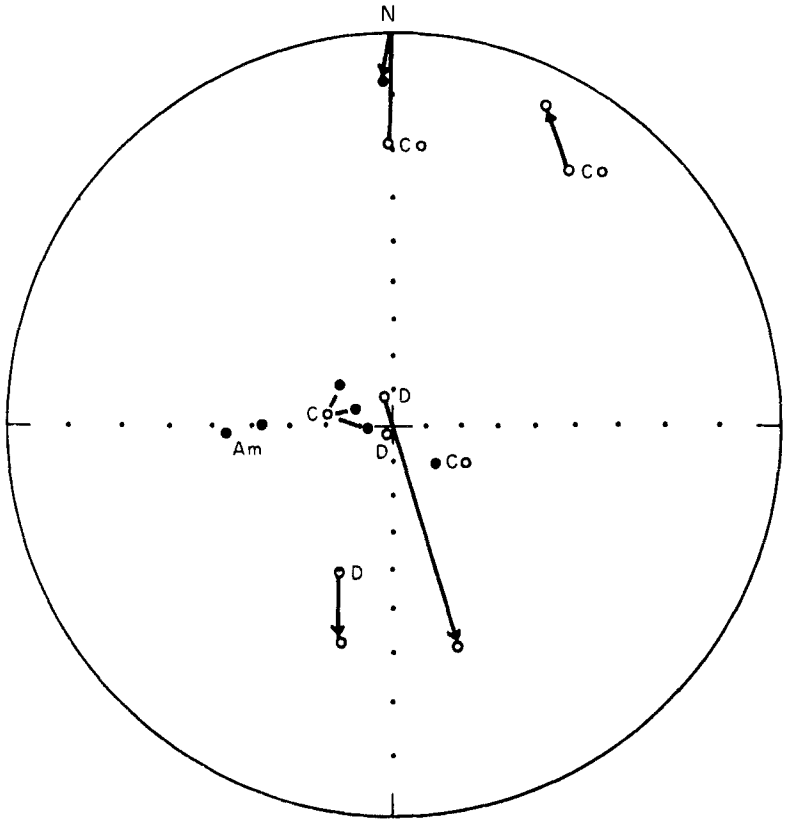


FIG. 4. Equal angle projections of the directions of magnetization of the North Wall Amphibolite Dyke (Am), the Central Dolerite Dyke (D) and the contact ore (Co). The arrowed lines connect points before and after thermal cleaning.

(c) *Iron Prince.* The Iron Prince ore body occupies a southward pitching syncline. It consists of several ore lenses separated by highly decomposed amphibolite. The ore is soft and powdery and in places the banding of the original jaspilite structure is clearly visible. The hematite is fine grained with very occasional inclusions of magnetite.

Thirty-one samples were collected from different quarry levels and the NRM directions are shown in Fig. 3. Only four samples have positive inclinations, all

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others are negatively magnetized. We shall denote them as the positive and negative groups of Iron Prince. Application of the χ^2-test shows the declination to be Fisherian but not the inclinations. Samples with positive inclination were not included in the computation of the mean.

The thermal and a.c. demagnetization characteristics of the samples are much the same as for Iron Monarch and only a few samples show a substantial change in direction, presumably due to the small inclusions of magnetite.

(d) *Iron Duke.* Deep drilling has revealed quite a complex structure of the Iron Duke ore body (Catley 1963). The high-grade hematite exposed near the surface changes into magnetite-carbonate at depth. The hematite ore itself frequently contains well developed crystals of magnetite.

The scatter in the NRM directions is rather large (Fig. 3) and does not decrease with demagnetization treatment. The changes in direction during demagnetization are quite large but not consistent. The thermal demagnetization curve shows that the remanence is less thermally stable than that of the other ore bodies and might be due to magnetite.

4. Intensity of magnetization

The intensity of the remanent magnetization lies between 10^{-4} and 10^{-2} e.m.u. cm^{-3}. This is several orders of magnitude lower than the TRM, which is characteristically of the order 10^{-1} e.m.u. cm^{-3} in a field of 0·6 oersted. As the NRM is extremely stable and should have large relaxation times, it seems unlikely that this is a TRM that has subsequently decayed. It appears unlikely therefore that the magnetization is TRM in origin, which precludes any ore-forming process which would have produced a TRM.

Laboratory experiments on the Chemical Remanent Magnetization of magnetite, produced by reduction of hematite (Haigh 1958; Kobayashi 1959), showed that the CRM has a lower intensity than the corresponding TRM. Experiments on the production of hematite from maghemite (Porath 1967) suggest that the same conclusion holds true for the CRM of hematite. We therefore suggest that the NRM of the hematite ore bodies is probably a CRM, which is in accordance with a replacement origin of the ore.

5. Magnetic anisotropy

The anisotropy of magnetic susceptibility was measured for the majority of samples using a low field torque meter as described by King & Rees (1962). A low field torque meter was used in preference to a high field torque meter for reasons discussed previously (Porath & Chamalaun 1966). The object of the measurements was to find out whether there existed a preferred crystalline alignment of hematite, which might be related to the stress pattern prevailing during the ore-forming process. Since the remanent vector lies within the basal plane of hematite, strong anisotropy would indicate that the palaeomagnetic directions were controlled by the anisotropy rather than the ambient field.

The distribution of the axes of maximum and minimum susceptibility are shown in Fig. 5. The scatter is quite large, but in the case of Iron Prince and Iron Duke there is a definite tendency for the maxima to be grouped along the strike of the banding planes and the minima to lie along the plane perpendicular to them. Iron Monarch, which shows hardly any remnants of the original banding planes, also does not show any preferred alignment of the susceptibility axes.

The agreement between the anisotropy pattern and the banding suggests that the textural anisotropy of the host rocks has been partially preserved. To obtain a textural anisotropy in the ore the specimens need to consist of layers of different

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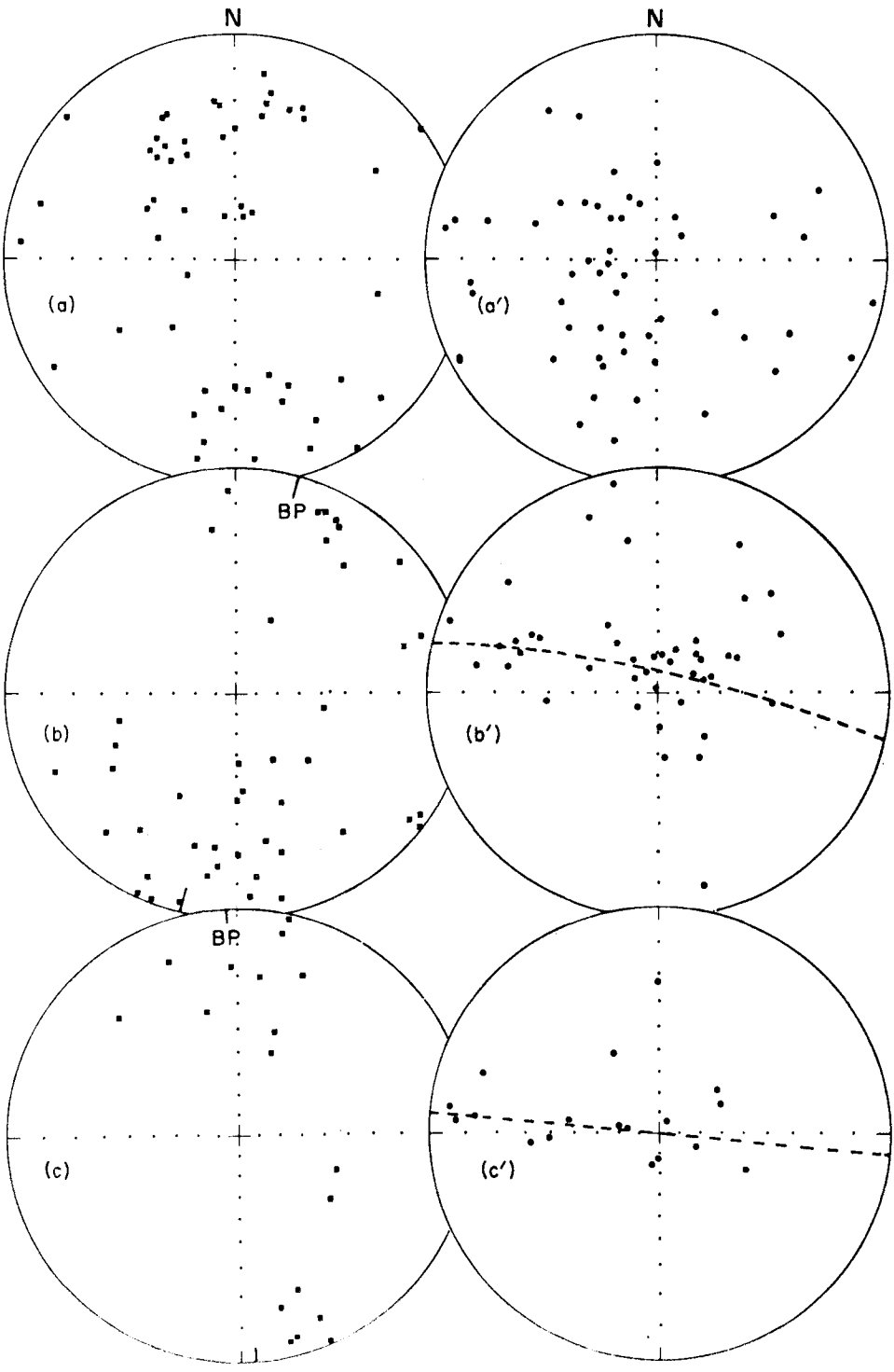


FIG. 5. Equal angle lower hemisphere projections of the distribution of the maximum (left) and minimum (right) susceptibility axes of Iron Monarch (a), Iron Prince (b) and Iron Duke (c). BP indicates the direction of strike of the banding planes, and the dashed lines join the poles of the banding planes.

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susceptibility. This could be produced by assuming layers 'rich' and 'poor' in magnetite corresponding to iron oxide and silica layers of the jaspilites before their replacement.

In general, the anisotropies are low and the magnetic lineations and foliations rarely exceed the 1 per cent. The susceptibility ellipsoids are mostly triaxial. The anisotropy is clearly too small to have had any effect upon the palaeomagnetic directions.

6. Discussion of the palaeomagnetic results

Of the ore bodies investigated Iron Monarch and Iron Prince show a consistent pattern of magnetization directions. The directions are given with respect to the present horizontal. As there is no geological evidence for tilting of the ore bodies after they were formed, no further corrections were made. Evidence for the stability of the remanent magnetization is provided by the thermal and AC demagnetization experiments, which showed that the magnetization is extremely stable and is unlikely to have been affected since its acquisition. Furthermore, the means of the three groups of directions are well away from the present field direction and the presence of two groups of negative and positive polarities in both ore bodies suggests in itself a certain degree of stability. If the NRM is a chemical remanent magnetization, it seems reasonable to assume that the magnetization was acquired during the ore-forming process. A comparison of the palaeomagnetic poles of the ores with the Australian polar wandering curve should then lead to an estimate of the age of the ore formation.

The interpretation of the results of Iron Monarch in terms of meaningful palaeomagnetic poles presents some difficulties. A satisfactory explanation should account for the scatter and somewhat elongated distribution of directions of the positive group of Iron Monarch and the fact that the negative and positive group give significantly different orientated dipole axes. If the positive and negative groups are interpreted to indicate polarity reversals of the geomagnetic field, then the fact that they are not antiparallel has to be accounted for. A divergence from antiparallelism is usually attributed to the presence of a secondary component of magnetization. However, demagnetization experiments do not show two components and there is no mineralogical evidence to support the suggestion that there were two stages of mineralization which might have yielded two magnetically equally stable hematite generations.

The basic difficulty that prevents a satisfactory interpretation to be formulated is that the time span covered by the samples and their stratigraphical sequence is unknown. Since positive samples from Iron Monarch do not show any obvious spatial distribution, it suggests that they occur as pockets within the ore body. This might be expected if a replacement origin of the ore is assumed. The replacement process would probably have started along structural weaknesses in different parts of the body and proceeded outwards from them. Consequently, adjacent samples could very well be different in age, and the mean direction of a sample could therefore be an average over quite a considerable time.

It is quite possible that a time span of several tens of millions of years is involved in the ore-forming process. If this is the case the sampling density is inadequate and each sample may represent readings of the geomagnetic field several millions of years apart. Hence the distribution of the directions and the scatter probably reflects polar wandering rather than secular variation. The fact that the positive and negative groups are not antiparallel can then be explained by arguing that they represent segments of the polar wandering curve before and after a polarity reversal. On this interpretation the palaeomagnetic poles computed from the positive and negative groups of Iron Monarch (see Table 1 and Fig. 6) represent the field during various stages of the ore-forming process.

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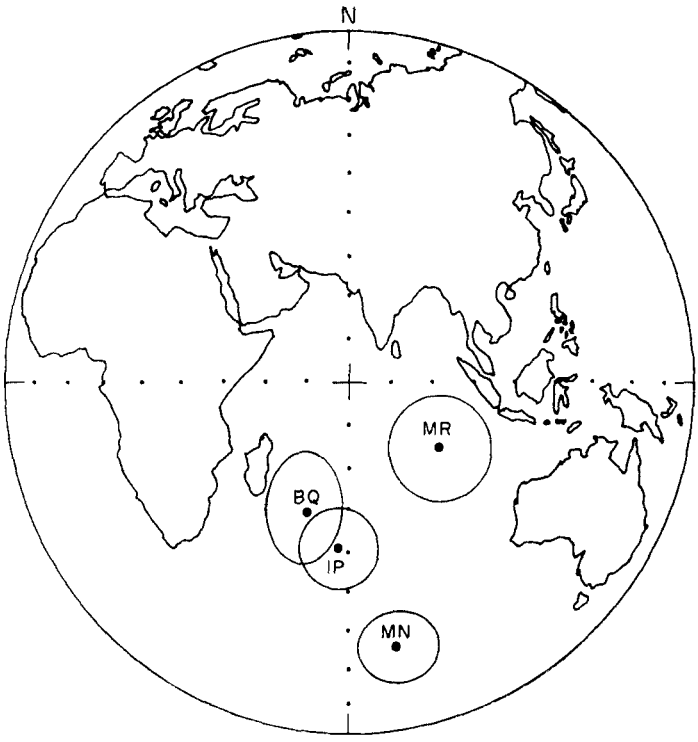


FIG. 6. Equal area projection of the palaeomagnetic poles relative to Australia corresponding to the Iron Monarch positive group (MR), the Iron Monarch negative group (MN), the Iron Prince (IP) and the Buldiva Quartzite (BQ).

The palaeomagnetic pole of the Iron Prince ore body, together with the mean direction and statistical parameters are listed in Table 1 and the pole is shown in Fig. 6. The non-Fisherian distribution of inclinations is difficult to explain on polar wandering alone, but may be the result of inadequate sampling.

Comparison of the poles of Iron Prince and Iron Monarch with other Australian poles for the Phanerozoic clearly suggests a Pre-Cambrian age for the magnetization. Unfortunately there are no well-dated Pre-Cambrian poles for Australia, but it may be noted that the pole of the Iron Prince agrees well with the Buldiva Quartzite pole determined by Irving & Green (1958), which is thought to be upper Proterozoic in age. When more Pre-Cambrian palaeomagnetic poles become available, it should be possible to establish a more precise age for the ore. In the meantime these pole positions should be considered as working-poles only.

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Department of Geophysics and Geochemistry,
The Australian National University,
Canberra.

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